

Lab 7: Error Debugging with AI:

Systematic approaches to finding and fixing bugs

2303A51682

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Assignment-7.3

Task 1: Fixing Syntax Errors

Function contains syntax error (missing colon).

Buggy Code `def add(a, b) return a + b`

AI Prompt Used

"Find the syntax error in this Python function and correct it." **Corrected Code** `def add(a, b):
return a + b
print(add(5, 3))`

Explanation :

- Python requires a **colon (:)** after function declaration.
- Without colon → `SyntaxError` occurs.
- AI detected missing colon and corrected it.

Expected Output :

8

Task 2: Debugging Logic Errors in Loops

Loop runs infinitely because increment is missing.

Buggy Code

```
i = 1 while i
<= 5:
    print(i)
```

AI Prompt Used

"Why is this loop infinite? Fix the logic error."

Corrected Code

```
i = 1 while i
<= 5:
    print(i)
    i += 1
```

Explanation

- Value of i never changes in buggy code.
- Condition $i \leq 5$ remains true forever.
- AI added increment $i += 1$.

Expected Output

```
1
2
3
4
```

Task 3: Handling Runtime Errors (Division by Zero)

Program crashes when divisor is zero.

Buggy Code

```
def divide(a, b):  
    return a / b  
print(divide(10, 0))
```

AI Prompt Used

"Identify runtime error and add safe exception handling."

Corrected Code

```
def divide(a, b):  
    try:  
        return a / b  
    except ZeroDivisionError:  
        return "Cannot divide by zero"  
print(divide(10, 0))
```

Explanation

- Division by zero raises **ZeroDivisionError**.
- AI added try-except block.
- Prevents program crash.

Expected Output

Cannot divide by zero

Task 4: Debugging Class Definition Errors

Constructor missing self parameter.

Buggy Code class Student:

```
def __init__(name, age):  
    name = name  
age = age
```

AI Prompt Used

"Fix constructor error and explain why self is required."

Corrected Code class Student:

```
def __init__(self, name, age):  
    self.name = name  
self.age = age  
s1 = Student("Sai", 19)  
print(s1.name, s1.age)
```

Explanation

- self represents current object instance.
- Without self, attributes are not stored inside object.

- AI corrected constructor and variable assignment.

Expected Output

Sai 19

Task 5: Resolving Index Errors in Lists

Program accesses invalid list index.

Buggy Code numbers

```
= [10, 20, 30]
```

```
print(numbers[5])
```

AI Prompt Used

"Identify index error and suggest safe list access."

Corrected Code (Method 1: Length Check) numbers

```
= [10, 20, 30]
```

```
index = 5 if index <
```

```
len(numbers):
```

```
print(numbers[index])
```

```
else:
```

```
    print("Index out of range")
```

Corrected Code (Method 2: Exception Handling)

```
numbers = [10, 20, 30]

try:
    print(numbers[5]) except
IndexError:
    print("Index out of range")
```

Explanation

- Accessing index greater than list size causes **IndexError**.
- AI suggested:
 - Bounds checking using len() ◦
 - OR try-except handling

Expected Output

Index out of range